

The running–coupling row, completed

Group I, Row V: $\mathfrak{A}^{(1)\dagger}\bar{\mathfrak{C}}^\dagger \times \mathfrak{B}_2$ with $\mathfrak{C}$ splitting the *measured* gluon — the z integration, the UV in $\overline{\text{MS}}$, and the p^+ integration

$$r = z - x, \quad s = y - x, \quad P = x' - w', \quad \xi = \frac{p^+}{k^+}, \quad \bar{\xi} = 1 - \xi, \quad q = \bar{\xi}k_\perp, \quad M^2 = \frac{\xi}{\bar{\xi}}s^2, \quad D = (r - s)^2 + M^2, \quad D_0 = \frac{s^2}{\xi}, \quad L = \log \frac{k^+}{\Lambda}$$

Verdict in one paragraph. Your note is structurally right and the hard part — recognising that the $r \rightarrow 0$ residue is $C_{\text{UV}}(\xi) = \xi\bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}(\xi)}{2N_c}$, and that $\int_{\Lambda/k^+}^{1-\Lambda/k^+} C_{\text{UV}} = 2L - \frac{11}{6}$ — is correct, and I confirm it to eight digits. Four things need fixing: **(i)** eq. (1.8) is missing a factor $\frac{1}{2}$; **(ii)** eq. (1.9) is missing an r^2 (it is dimensionally inconsistent as written); **(iii)** the overall $\frac{p^+}{k^+}$ in (1.1) is a duplication and the overall sign is opposite — your own (1.11) already has it right; **(iv)** the $\frac{1}{\epsilon}$ of (1.19)–(1.20) is $-\frac{1}{\epsilon}$ in your own $d = 2 - 2\epsilon$ convention, i.e. $\frac{1}{\epsilon_{\text{UV}}}$ with $d = 2 + 2\epsilon_{\text{UV}}$. Two things are *missing*: a finite tensor term that the replacement $\frac{(P \cdot r)(s \cdot r)}{P^2 r^4} \rightarrow \frac{s \cdot P}{d_\perp P^2 r^2}$ throws away, and — more importantly — the remainder $\mathcal{I}(\xi)$ is **not** uniformly finite at $\xi \rightarrow 1$: it carries its own $\frac{1}{\xi} \log \bar{\xi}$, which **halves** the $\log^2 \frac{k^+}{\Lambda}$ of your (1.22). And the plus prescription works at $\xi \rightarrow 0$ but **fails** at $\xi \rightarrow 1$.

1. What I checked, line by line

Line of your note	Status	What it should be
(1.1), the six–group bracket	correct	reproduces $\mathfrak{C}_{1jki} \mathfrak{B}_2^{jk}$ to 10^{-12} on 400 random configurations
(1.1), the prefactor	wrong	drop the overall $\frac{p^+}{k^+}$ (it is already inside the bracket as $\xi\bar{\xi}$) and flip the sign
(1.2) denominators, (1.3) phase	correct	but state $q = \bar{\xi}k_\perp$ explicitly — the ξ is what makes $\log q^2 = \log(\bar{\xi}^2 k^2)$
(1.4), (1.5), (1.6) $\rightarrow$ (1.7)	correct	—
(1.8)	wrong	$\left[1 - \frac{r \cdot s}{s^2}\right] \longrightarrow \left[1 - \frac{r \cdot s}{2s^2}\right]$
(1.9)	wrong	$\frac{s^2 - s \cdot r}{2(s - r)^2} \longrightarrow \frac{s^2 - s \cdot r}{2r^2(s - r)^2}$
(1.10)	correct	—
(1.14), (1.15) — the UV residue	correct	$\frac{s \cdot P}{P^2 r^2} C_{\text{UV}}(\xi)$; confirmed numerically to 8 digits
(1.16)–(1.18)	unsafe route	you used the $d = 2$ Fourier transform $\frac{2\pi i q^i}{q^2}$ inside a d –dimensional convolution and then set two scaleless integrals to zero — that trades a UV pole for an IR one
(1.19)–(1.20)	sign of pole	$\pi \left[\frac{1}{\epsilon} + \log \frac{\bar{\mu}^2}{q^2}\right] \rightarrow \pi \left[-\frac{1}{\epsilon} + \log \frac{\bar{\mu}^2}{q^2}\right]$ in $d = 2 - 2\epsilon$
(1.22) — the $\frac{1}{\epsilon}$ line	correct	$2L - \frac{11}{6}$, verified
(1.22) — the $\log^2$ line	incomplete	$\log^2 L \rightarrow \frac{1}{2} \log^2 L$ once $\mathcal{I}(\xi)$ ’s own $\frac{1}{\xi} \log \bar{\xi}$ is included (§7)

2. The corrected starting point

Put x at the origin so that $r = z$, $s = y$. With the $\mathfrak{C}$ delta function $\delta^{(2)}[(w - z) + \frac{p^+}{k^+}(z - x)]$ solved for $w = z - \xi r$ (Jacobian 1), the measurement phase is

$$e^{-ik \cdot (w' - z)} e^{-ik \cdot (z - x)} = e^{-ik \cdot w'} e^{ik \cdot x} e^{i\bar{\xi} k \cdot r} \implies \boxed{q = \bar{\xi} k_\perp}$$

and the row reads

$$\left. \frac{d^3 N_2^{\text{NLO}}}{d^3 k} \right|_{\text{V}} = \frac{-1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_{x,y,x',w'} e^{-ik \cdot (w' - x)} \int_{\Lambda/k^+}^{1-\Lambda/k^+} \frac{d\xi}{2\pi} \int d^d r e^{iq \cdot r} \frac{T(r)}{\xi D} \times \mathcal{U}$$

with $\mathcal{U} = [U^{ab'}(x') - U^{ab'}(w')] [-N_c U^{af}(x) + N_c U^{af}(y)] \rho^{b'}(x') \rho^f(y)$. This is exactly your (1.11) — i.e. *your (1.11) is right and your (1.1) is not*; they differ by the stray $\frac{p^+}{k^+}$. The overall minus comes from $\frac{2}{(2\pi)^3} \mathfrak{A}^{(1)\dagger} \bar{\mathfrak{C}}^\dagger \mathfrak{B}_2 = -\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{p^+(k^+ - p^+)}{k^{+3} D}$; check it against your sign conventions for $\bar{\mathfrak{A}}, \bar{\mathfrak{B}}$, since that is where it lives.

The corrected $T(r)$

$$T(r) = \frac{d_\perp \xi \bar{\xi}}{2} \left[\frac{2(P \cdot r)(s \cdot r)}{P^2 r^4} - \frac{P \cdot r}{P^2 r^2} \right] \quad (1.4)$$

$$+ \frac{\xi}{\bar{\xi}} \left[\frac{P \cdot s}{P^2 r^2} + \frac{(P \cdot s)(r \cdot s - r^2)}{2P^2 r^2 (s - r)^2} \right] \quad (1.5)$$

$$+ \bar{\xi} \frac{P \cdot r}{P^2 r^2} \left[2 - \frac{2(s \cdot r)}{r^2} - \frac{s \cdot r}{2s^2} \right] \quad (1.7)$$

$$+ \frac{\xi}{\bar{\xi}} \frac{P \cdot s - P \cdot r}{P^2 r^2} \left[1 - \frac{r \cdot s}{2s^2} \right] \quad (1.8), \text{ corrected}$$

$$- \xi \frac{P \cdot r}{P^2} \left[\frac{2(s \cdot r)}{r^4} + \frac{s^2 - s \cdot r}{2r^2(s - r)^2} - \frac{1}{2r^2} \right] \quad (1.9), \text{ corrected}$$

$$+ \left[\frac{2(P \cdot r)(r \cdot s)}{P^2 r^4} - \frac{P \cdot r}{P^2 r^2} + \frac{P \cdot s}{2P^2 s^2} - \frac{(P \cdot s)(r \cdot s)}{2P^2 r^2 s^2} + \frac{(r \cdot s)[P \cdot s - P \cdot r]}{2P^2 r^2 (s - r)^2} \right] \quad (1.10)$$

How the two errors were found. Each of the six groups of (1.1) was contracted with $\frac{P_i r^m}{P^2 r^2}$ numerically and compared with your corresponding line, over 400 random (r, s, P, ξ) . Worst discrepancies as written: (1.8) 2.5×10^1 , (1.9) 1.1×10^1 ; all four other lines $\leq 1.9 \times 10^{-12}$. With the two fixes above, all six agree to $\leq 5 \times 10^{-12}$. The (1.9) error is also visible by inspection: $\frac{s^2 - s \cdot r}{2(s - r)^2}$ is dimensionless while its two neighbours inside the same bracket are $1/\text{length}^2$.

3. Where, exactly, the r integral can diverge

Before integrating, locate every dangerous point. There are three.

(a) $r \rightarrow 0$, i.e. $z \rightarrow x$ — this is your UV. Every term of T that survives the angular average at small r behaves as $1/r^2$, and $\frac{1}{\xi D} \rightarrow \frac{1}{\xi D_0} = \frac{1}{s^2}$. In two dimensions $\int_0 \frac{d^2 r}{r^2} = 2\pi \int_0 \frac{d\rho}{\rho}$ is log divergent. This is a genuine short-distance divergence and is what §4–§6 regulate.

(b) $r \rightarrow s$, i.e. $z \rightarrow y$ — harmless. Three terms carry $\frac{1}{(s-r)^2}$, and in every one of them the numerator vanishes linearly there:

$$r \cdot s - r^2 = r \cdot (s - r), \quad s^2 - s \cdot r = s \cdot (s - r), \quad P \cdot s - P \cdot r = P \cdot (s - r).$$

So each behaves as $1/|s - r|$, and $\int d^2 r |s - r|^{-1}$ converges. Note also that D never vanishes: $D = (r - s)^2 + M^2$ with $M^2 = \frac{\xi}{\bar{\xi}} s^2 > 0$. This is why the $\xi \rightarrow 0$ endpoint, where $M^2 \rightarrow 0$, does not produce a second collinear pole.

(c) $r \rightarrow \infty$ — a real logarithm, and your note misses it. Expand T at large r . The $\frac{1}{r^2}$ pieces all decay, but the third entry of (1.10),

$$\frac{P \cdot s}{2P^2 s^2},$$

is r -independent. Hence $T(r) \rightarrow \frac{P \cdot s}{2P^2 s^2} + O(1/r)$ and $\frac{T}{\xi D} \rightarrow \frac{1}{\xi r^2} \frac{P \cdot s}{2P^2 s^2}$, giving another logarithm, cut off only by the phase at $r \sim 1/q = 1/(\bar{\xi}|k_\perp|)$. At generic ξ this is a finite number; at $\xi \rightarrow 1$ it is precisely what halves your $\log^2$ (§7).

4. Isolating the UV: the angular average at $r \rightarrow 0$

Only even-in- r structures survive the $\hat{r}$ average, and only those scaling as $1/r^2$ diverge. Going through the six lines:

- (1.4): the first entry is $\frac{d_\perp \xi \bar{\xi} (P \cdot r)(s \cdot r)}{P^2 r^4}$ — even, $\sim r^{-2}$: **keep**. The second, $\propto \frac{P \cdot r}{r^2} \sim r^{-1}$, is odd: drops.
- (1.5): $\frac{\xi}{\bar{\xi}} \frac{P \cdot s}{P^2 r^2}$ — **keep**. The second entry $\rightarrow \frac{(P \cdot s)(r \cdot s)}{2P^2 r^2 s^2} \sim r^{-1}$, odd: drops.
- (1.7): $2\bar{\xi} \frac{P \cdot r}{P^2 r^2}$ is odd; the remaining two are finite. Nothing.
- (1.8): $\frac{\xi}{\bar{\xi}} \frac{P \cdot s}{P^2 r^2}$ — **keep**; $-\frac{\xi}{\bar{\xi}} \frac{P \cdot r}{P^2 r^2}$ odd; the rest finite.
- (1.9): every entry is odd or finite at $r \rightarrow 0$ (use $\frac{s^2 - s \cdot r}{2r^2(s - r)^2} \rightarrow \frac{1}{2r^2}$, which cancels against $-\frac{1}{2r^2}$). Nothing.
- (1.10): $\frac{2(P \cdot r)(r \cdot s)}{P^2 r^4}$ and $\frac{(r \cdot s)(P \cdot s)}{2P^2 r^2 s^2}$ from the last entry cancel against $-\frac{(P \cdot s)(r \cdot s)}{2P^2 r^2 s^2}$; what is left is odd or finite. Nothing.

$$T(r) \xrightarrow{r \rightarrow 0} T_{\text{UV}}(r) = d_{\perp} \xi \bar{\xi} \frac{(P \cdot r)(s \cdot r)}{P^2 r^4} + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2 r^2}$$

and, after $\langle \hat{r}^i \hat{r}^m \rangle = \frac{\delta^{im}}{d_{\perp}}$ (which cancels the $d_{\perp}$ exactly),

$$\langle T \rangle_{\hat{r}} \rightarrow \frac{1}{r^2} \frac{P \cdot s}{P^2} \underbrace{\left[\xi \bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right]}_{C_{\text{UV}}(\xi) = \frac{P_{gg}(\xi)}{2N_c}}.$$

This is your (1.15), and it is **right**.

Numerical confirmation — the angular average of $\rho^2 T / (\xi D)$ against $C_{\text{UV}}(\xi) \frac{P \cdot s}{P^2 s^2}$, at $s = (0.7, -0.4)$, $P = (-0.3, 0.9)$:

ξ	$\rho = 10^{-2}$	10^{-3}	10^{-4}	predicted
0.10	-8.96488340	-8.96518217	-8.96518516	-8.96518519
0.25	-3.43050462	-3.43055505	-3.43055555	-3.43055556
0.50	-2.19233580	-2.19230797	-2.19230770	-2.19230769
0.75	-3.43059049	-3.43055590	-3.43055556	-3.43055556
0.90	-8.96520571	-8.96518539	-8.96518519	-8.96518519

The symmetry under $\xi \leftrightarrow \bar{\xi}$ is exact, as it must be for $g \rightarrow gg$.

5. The two Fourier masters in $d = 2 - 2\epsilon$, from scratch

Start from the one identity everything else follows from. For $0 < \alpha < d/2$,

$$\int d^d r \frac{e^{iq \cdot r}}{(r^2)^\alpha} = \pi^{d/2} 2^{d-2\alpha} \frac{\Gamma(\frac{d}{2} - \alpha)}{\Gamma(\alpha)} (q^2)^{\alpha-d/2}.$$

Set $\alpha = 1$, $d = 2 - 2\epsilon$, and restore μ by the factor $\mu^{-2\epsilon}$ that makes the left side dimensionless:

$$\mu^{-2\epsilon} \int d^d r \frac{e^{iq \cdot r}}{r^2} = \pi^{1-\epsilon} 2^{-2\epsilon} \Gamma(-\epsilon) \left(\frac{q^2}{\mu^2} \right)^\epsilon = -\pi \left[\frac{1}{\epsilon} + \gamma_E + \log \frac{q^2}{4\pi\mu^2} \right] + O(\epsilon) \equiv -\pi \mathcal{L}_q,$$

$$\mathcal{L}_q \equiv \frac{1}{\epsilon} + \log \frac{q^2}{\bar{\mu}^2}, \quad \bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_E}.$$

For the tensor master, differentiate rather than integrate. First, exactly in d dimensions,

$$\partial^m \frac{r^i}{r^2} = \frac{\delta^{im}}{r^2} - \frac{2r^i r^m}{r^4} \implies \frac{r^i r^m}{r^4} = \frac{1}{2} \left[\frac{\delta^{im}}{r^2} - \partial^m \frac{r^i}{r^2} \right].$$

Next, from $\partial_{q_i} e^{iq \cdot r} = i r^i e^{iq \cdot r}$ applied to the scalar master,

$$\int d^d r e^{iq \cdot r} \frac{r^i}{r^2} = -i \partial_{q_i} \left[C_1 (q^2)^\epsilon \right] = -2i\epsilon C_1 q^i (q^2)^{\epsilon-1} \xrightarrow{\epsilon \rightarrow 0} \frac{2\pi i q^i}{q^2},$$

which is finite (the pole is killed by $\epsilon \Gamma(-\epsilon) = -\Gamma(1 - \epsilon)$). Since $\int e^{iq \cdot r} \partial^m g = -i q^m \int e^{iq \cdot r} g$,

$$\int d^d r e^{iq \cdot r} \partial^m \frac{r^i}{r^2} = -i q^m \cdot \frac{2\pi i q^i}{q^2} = \frac{2\pi q^i q^m}{q^2},$$

$$\mu^{-2\epsilon} \int d^d r \frac{e^{iq \cdot r}}{r^2} = -\pi \mathcal{L}_q, \quad \mu^{-2\epsilon} \int d^d r e^{iq \cdot r} \frac{r^i r^m}{r^4} = -\frac{\pi}{2} \mathcal{L}_q \delta^{im} - \pi \hat{q}^i \hat{q}^m.$$

The $d_\perp$ trap — do not set $d_\perp = 2$ before contracting. Trace the second master in d dimensions: $-\frac{\pi}{2}\mathcal{L}_q d - \pi = -\pi\mathcal{L}_q + \pi(\epsilon\mathcal{L}_q) - \pi = -\pi\mathcal{L}_q$, using $\epsilon\mathcal{L}_q \rightarrow 1$. It reproduces the scalar master *only* because the -2ϵ of $d_\perp$ eats the pole and leaves a finite $+\pi$. Your replacement $\frac{(P \cdot r)(s \cdot r)}{P^2 r^4} \rightarrow \frac{s \cdot P}{d_\perp P^2 r^2}$ (just above your 1.15) is correct for the pole and the log, but it silently discards that $+\pi$ and the whole $\hat{q}^i \hat{q}^m$ structure — which is not proportional to $P \cdot s$ at all.

Cutoff cross-check (hard cutoff $|r| > \rho_0$, $L_\rho = \log \frac{2}{q\rho_0} - \gamma_E$, at $q = 1.3$, $\rho_0 = 10^{-6}$, by oscillatory quadrature):

quantity	numerical	closed form
$\int_{ r >\rho_0} d^2 r e^{iqr} / r^2$	85.8853488532	$2\pi L_\rho = 85.8853488532$
δ^{im} part of $\int e^{iqr} r^i r^m / r^4$	44.5134707534	$\pi(L_\rho + \frac{1}{2}) = 44.5134707534$
$\hat{q}^i \hat{q}^m$ part	-3.14159265359	$-\pi$

The cutoff scheme puts the extra $\frac{\pi}{2}\delta^{im}$ in a different place from $\overline{\text{MS}}$ (compare $\pi(L_\rho + \frac{1}{2})$ with $-\frac{\pi}{2}\mathcal{L}_q \leftrightarrow \pi L_\rho$), but the traces agree and — as §6 shows — so does the physical finite remainder.

Why your (1.16)–(1.18) flips the pole. The convolution identity itself is fine: $\int d^2 r \frac{e^{iqr}}{r^2} = (2\pi)^2 \int \frac{d^2 l}{(2\pi)^2} \frac{l \cdot (l-q)}{l^2 (l-q)^2}$. But in d dimensions $\int d^d r e^{iqr} \frac{r^i}{r^2}$ is $2i\pi^{1-\epsilon} 2^{-2\epsilon} \Gamma(1-\epsilon) \frac{q^i}{(q^2)^{1-\epsilon}}$, not $\frac{2\pi i q^i}{q^2}$; the ϵ -dependent factors multiply a $\frac{1}{\epsilon}$ and therefore matter. On top of that, setting $\int \frac{d^d l}{l^2} = \int \frac{d^d l}{(l-q)^2} = 0$ is the statement $\frac{1}{\epsilon_{UV}} - \frac{1}{\epsilon_{IR}} = 0$, which is exactly the bookkeeping you are trying to track. The net effect is a pole of the wrong sign. A one-line sanity check settles it: with a cutoff, $\int_{\rho_0} d^2 r \frac{e^{iqr}}{r^2} = 2\pi \log \frac{2e^{-\gamma_E}}{q\rho_0}$ is *large and positive* as $\rho_0 \rightarrow 0$, so the pole must enter with the *same* sign as $\log \frac{\bar{\mu}^2}{q^2}$:

$$\mu^{-2\epsilon} \int d^d r \frac{e^{iqr}}{r^2} = \pi \left[-\frac{1}{\epsilon} + \log \frac{\bar{\mu}^2}{q^2} \right] \equiv \pi \left[\frac{1}{\epsilon_{UV}} + \log \frac{\bar{\mu}^2}{q^2} \right], \quad \frac{1}{\epsilon_{UV}} \equiv -\frac{1}{\epsilon}.$$

So your expression (1.20) is the right one provided ϵ there means ϵ_{UV} , i.e. $d = 2 + 2\epsilon_{UV}$ — which is the correct sign of the analytic continuation for a position-space short-distance divergence anyway. Just say so, or the $\frac{1}{\epsilon}$ will fight the log when you combine rows.

6. The r integration: subtract, integrate exactly, keep the rest finite

Your split $\frac{1}{D} = \frac{1}{D_0} + \left(\frac{1}{D} - \frac{1}{D_0}\right)$ is the right *idea*, but it only softens the $\frac{1}{r^2 D}$ term; the $\frac{1}{r^4}$ tensor term is still divergent. Subtract the whole $r \rightarrow 0$ asymptotics at once — nothing else changes and every step stays exact:

$$\int d^d r e^{iq \cdot r} \frac{T(r)}{\xi \bar{D}(r)} = \underbrace{\frac{1}{s^2} \int d^d r e^{iq \cdot r} T_{UV}(r)}_{\text{carries the whole pole; done exactly}} + \underbrace{\int d^2 r e^{iq \cdot r} \left[\frac{T(r)}{\xi \bar{D}(r)} - \frac{T_{UV}(r)}{s^2} \right]}_{\equiv \mathcal{I}(\xi), \text{ finite at } \epsilon = 0}$$

(using $\bar{\xi} D \rightarrow \bar{\xi} D_0 = s^2$ at $r = 0$). The subtraction works because, by construction, the bracket has no r^{-2} even part left; what survives at small r is $O(1/r)$ and odd, which is integrable in two dimensions. Numerically, $\rho^2 \langle \cdot \rangle_{\hat{r}}$ for the subtracted integrand falls linearly in ρ , as it must:

ξ	$\rho = 10^{-2}$	10^{-3}	10^{-4}
0.25	$+5.09 \times 10^{-5}$	$+5.09 \times 10^{-7}$	$+5.09 \times 10^{-9}$
0.50	-2.81×10^{-5}	-2.81×10^{-7}	-2.81×10^{-9}
0.75	-3.49×10^{-5}	-3.49×10^{-7}	-3.49×10^{-9}

The exact piece

Feed T_{UV} to the two masters, keeping $d_\perp = d = 2 - 2\epsilon$ until the very end:

$$\begin{aligned} \int d^d r e^{iqr} T_{UV} &= d \xi \bar{\xi} \frac{P^i s^m}{P^2} \left[-\frac{\pi}{2} \mathcal{L}_q \delta^{im} - \pi \hat{q}^i \hat{q}^m \right] + \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2} (-\pi \mathcal{L}_q) \\ &= \xi \bar{\xi} \left[-\pi \mathcal{L}_q (1 - \epsilon) \frac{P \cdot s}{P^2} - 2\pi (1 - \epsilon) \frac{(P \cdot \hat{q})(s \cdot \hat{q})}{P^2} \right] - \pi \mathcal{L}_q \left(\frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} \right) \frac{P \cdot s}{P^2} \\ &= -\pi \mathcal{L}_q C_{UV}(\xi) \frac{P \cdot s}{P^2} + \pi \xi \bar{\xi} \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2}, \end{aligned}$$

where $\epsilon \mathcal{L}_q \rightarrow 1$ generated the $+\pi \xi \bar{\xi} \frac{P \cdot s}{P^2}$, and $\hat{q} = \hat{k}$ because $q = \bar{\xi} k$. Exactly the same finite leftover comes out of the cutoff scheme $(2\xi \bar{\xi} [\pi(L_\rho + \frac{1}{2})(P \cdot s) - \pi(P \cdot \hat{q})(s \cdot \hat{q})]/P^2)$, which is the check that the $\frac{1}{2}$ is real and not a scheme artefact.

Result at fixed ξ , in $\overline{\text{MS}}$:

$$\int d^d r e^{iq \cdot r} \frac{T(r)}{\xi D} = \frac{\pi}{s^2} C_{\text{UV}}(\xi) \frac{P \cdot s}{P^2} \left[\frac{1}{\epsilon_{\text{UV}}} + \log \frac{\bar{\mu}^2}{\xi^2 k^2} \right] + \frac{\pi}{s^2} \xi \bar{\xi} \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2} + \mathcal{I}(\xi).$$

The middle term is new — it is what the $d_\perp$ replacement lost.

7. The remainder $\mathcal{I}(\xi)$ is *not* uniformly finite — and this halves your $\log^2$

Your (1.21) says “ $\mathcal{I}(\xi)$ is a finite term” and then integrates only the explicit $\log \frac{\bar{\mu}^2}{\xi^2 k^2}$ over ξ . $\mathcal{I}(\xi)$ is finite at every fixed ξ , but it blows up at the $\xi \rightarrow 1$ endpoint faster than $\frac{1}{\xi}$, and the ξ integral sees that.

The $\bar{\xi} \rightarrow 0$ end, analytically

Only one line of T is $\frac{1}{\xi}$ -enhanced, namely (1.5), with $g(r) = \frac{1}{r^2} + \frac{r \cdot (s-r)}{2r^2(s-r)^2}$; and one line tends to a constant, namely the $\frac{P \cdot s}{2P^2 s^2}$ of (1.10). Note that $\bar{\xi} D = \bar{\xi}(r-s)^2 + \xi s^2$, so for $r \gg s$, with $u \equiv \bar{\xi} r^2/s^2$ and $g \rightarrow \frac{1}{2r^2}$,

$$\frac{T}{\bar{\xi} D} - \frac{T_{\text{UV}}}{s^2} \rightarrow \frac{P \cdot s}{P^2 s^2} \left\{ \frac{1}{1+u} \left[\frac{1}{2u} + \frac{1}{2} \right] - \frac{1}{u} \right\} \frac{1}{s^2} \cdot s^2 = -\frac{P \cdot s}{2P^2 s^2} \frac{1}{\bar{\xi} r^2},$$

the u -dependence cancelling completely. Hence, with the phase cutting at $r \sim 1/q = 1/(\bar{\xi}|k_\perp|)$ and the expansion starting at $r \sim s$,

$$\mathcal{I}(\xi) \xrightarrow{\bar{\xi} \rightarrow 0} -\frac{\pi}{\bar{\xi}} \frac{P \cdot s}{P^2 s^2} \log \frac{1}{\bar{\xi} |k_\perp| s} + \text{const} \implies \bar{\xi} \mathcal{I} \simeq -\pi \frac{P \cdot s}{P^2 s^2} \log \frac{1}{\bar{\xi}}.$$

Numerically (2D quadrature inside $|r| < R_0$ plus the closed-form tail above, at the same s, P and $k = (1, 0)$; $-\pi \frac{P \cdot s}{P^2 s^2} = 3.061039$):

$\bar{\xi}$	$\bar{\xi} \text{Re } \mathcal{I}$	local slope in $\log \frac{1}{\bar{\xi}}$
3×10^{-2}	11.563648	—
1×10^{-2}	15.039494	3.163851
3×10^{-3}	18.781176	3.107779
1×10^{-3}	22.154936	3.070929
3×10^{-4}	25.839193	3.060084
1×10^{-4}	29.238683	3.094349

The $\xi \rightarrow 0$ end — clean

There $q = \bar{\xi} k \rightarrow k$, so the explicit $\log \frac{\bar{\mu}^2}{q^2}$ stays finite, and $M^2 \rightarrow 0$ does not bite because every $\frac{1}{(s-r)^2}$ numerator vanishes at $r = s$ (§3b). Numerically $\xi \mathcal{I}$ tends to a constant, slope $\rightarrow 0$:

ξ	10^{-2}	3×10^{-3}	10^{-3}	3×10^{-4}	10^{-4}
$\xi \text{Re } \mathcal{I}$	2.601994	2.611887	2.614347	2.615165	2.615393
slope	—	0.0082	0.0022	0.00068	0.00021

Consequence. Near $\bar{\xi} = 0$ the explicit log contributes $\pi C_{\text{UV}} \frac{P \cdot s}{P^2 s^2} \cdot 2 \log \frac{1}{\bar{\xi}} \rightarrow \frac{2\pi}{\bar{\xi}} \frac{P \cdot s}{P^2 s^2} \log \frac{1}{\bar{\xi}}$, while $\mathcal{I}$ contributes $-\frac{\pi}{\bar{\xi}} \frac{P \cdot s}{P^2 s^2} \log \frac{1}{\bar{\xi}}$. The sum is **half** the first. Since $\int_\lambda \frac{d\bar{\xi}}{\bar{\xi}} \log \frac{1}{\bar{\xi}} = \frac{1}{2} \log^2 \frac{1}{\lambda}$, the $\log^2 \frac{k^+}{\Lambda}$ of your (1.22) becomes $\frac{1}{2} \log^2 \frac{k^+}{\Lambda}$. The $\frac{1}{\epsilon}$ line of (1.22) is untouched, because the pole’s residue is exactly $C_{\text{UV}}(\xi)$ with no further ξ dependence.

8. The p^+ integration, step by step

With $\lambda = \Lambda/k^+$ and $L = \log \frac{k^+}{\Lambda} = \log \frac{1}{\lambda}$. Three elementary integrals.

(i) *The pole, and* $\log \frac{\bar{\mu}^2}{k^2}$

$$\int_{\lambda}^{1-\lambda} d\xi \frac{\bar{\xi}}{\xi} = \log \frac{1-\lambda}{\lambda} - (1-2\lambda) \rightarrow L-1, \quad \int_{\lambda}^{1-\lambda} d\xi \frac{\xi}{\bar{\xi}} \rightarrow L-1, \quad \int_0^1 d\xi \xi \bar{\xi} = \frac{1}{6},$$

$$\boxed{\int_{\lambda}^{1-\lambda} d\xi C_{UV}(\xi) = 2L - 2 + \frac{1}{6} = 2L - \frac{11}{6}}$$

— your (1.22), confirmed. (Numerically at $\lambda = 10^{-6}$: 25.79768978 against 25.79768778.)

(ii) *The* $\log \frac{1}{\xi^2}$ *part of the explicit log*

Split $\log \frac{\bar{\mu}^2}{\xi^2 k^2} = \log \frac{\bar{\mu}^2}{k^2} + 2 \log \frac{1}{\xi}$. Term by term, with $u = \bar{\xi}$ where convenient:

$$\begin{aligned} -2 \int_{\lambda}^{1-\lambda} \frac{\xi}{\bar{\xi}} \log \bar{\xi} d\xi &= -2 \int_{\lambda}^{1-\lambda} \left(\frac{\log u}{u} - \log u \right) du = -2 \left[\frac{\log^2 u}{2} - u \log u + u \right]_{\lambda}^{1-\lambda} \rightarrow L^2 - 2, \\ -2 \int_0^1 \frac{\bar{\xi}}{\xi} \log \bar{\xi} d\xi &= -2 \left[\underbrace{\int_0^1 \frac{\log(1-\xi)}{\xi} d\xi}_{-\pi^2/6} - \underbrace{\int_0^1 \log(1-\xi) d\xi}_{-1} \right] = \frac{\pi^2}{3} - 2, \\ -2 \int_0^1 \xi \bar{\xi} \log \bar{\xi} d\xi &= -2 \int_0^1 (u - u^2) \log u du = -2 \left(-\frac{1}{4} + \frac{1}{9} \right) = \frac{5}{18}. \\ \Rightarrow -2 \int_{\lambda}^{1-\lambda} C_{UV}(\xi) \log \bar{\xi} d\xi &= L^2 + \frac{\pi^2}{3} - 4 + \frac{5}{18} = L^2 + \frac{\pi^2}{3} - \frac{67}{18}. \end{aligned}$$

Again exactly your (1.22). (At $\lambda = 10^{-6}$: 190.4360055 against 190.4359779.)

(iii) *The remainder*

$$\int_{\lambda}^{1-\lambda} \mathcal{I}(\xi) d\xi = \underbrace{-\frac{\pi}{2} \frac{P \cdot s}{P^2 s^2} L^2}_{\text{from } \frac{1}{\xi} \log \bar{\xi}} + \underbrace{(c_0 + c_1) L}_{\text{from the } \frac{a_0}{\xi}, \frac{c_1}{\xi} \text{ ends}} + \text{finite},$$

with c_0, c_1 depending on $|k_{\perp}|s, \hat{k} \cdot \hat{s}, \hat{k} \cdot \hat{P}$ (for the test point above, $c_0 = 2.61539$). These are ordinary numbers, not new divergences — but they do enter the single-log coefficient, so they must be computed alongside the other rows of Group I rather than dropped.

9. The final result for this row

Collecting, and using $\int \frac{d\xi}{2\pi}$ with the π from the masters (which is what turns your $\frac{q^4}{8\pi^4}$ into $\frac{q^4}{16\pi^4}$, correctly), and $\frac{P \cdot s}{P^2 s^2} = \frac{(x'-w') \cdot (y-x)}{(x'-w')^2 (y-x)^2}$.

$$\begin{aligned} \left. \frac{d^3 N_2^{\text{NLO}}}{d^3 k} \right|_V &= \frac{-1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{k^+} \int_{x,y,x',w'} e^{-ik \cdot (w'-x)} \left\{ \frac{(x'-w') \cdot (y-x)}{(x'-w')^2 (y-x)^2} \left[\left(\frac{1}{\epsilon_{UV}} + \log \frac{\bar{\mu}^2}{k_{\perp}^2} \right) \left(2L - \frac{11}{6} \right) \right. \right. \\ &\quad \left. \left. + \frac{1}{2} L^2 + \frac{\pi^2}{3} - \frac{67}{18} + (c_0 + c_1) L + \text{fin.} \right] \right. \\ &\quad \left. + \frac{1}{6} \frac{(x'-w') \cdot (y-x) - 2[(x'-w') \cdot \hat{k}][(y-x) \cdot \hat{k}]}{(x'-w')^2 (y-x)^2} \right\} \times \mathcal{U} \end{aligned}$$

Differences from your (1.22), in one list: overall sign; $\frac{1}{\epsilon} \rightarrow \frac{1}{\epsilon_{UV}}$; $\log^2 \rightarrow \frac{1}{2} \log^2$; the extra $\frac{1}{6}$ -weighted tensor term on the last line; and the extra $(c_0 + c_1)L$ in the single-log coefficient.

10. Can you use the plus prescription for the p^+ integral?

The test is always the same one. Writing the ξ integrand near an endpoint as $\frac{g(\xi)}{\xi}$,

$$\int_{\lambda}^1 \frac{d\xi}{\xi} g(\xi) = \int_0^1 d\xi \left[\frac{1}{\xi} \right]_+ g(\xi) + g(0) \log \frac{1}{\lambda}$$

is legitimate **if and only if** $g(0)$ is finite. Apply it at both ends.

At $\xi \rightarrow 0$ — yes. There $C_{UV} \rightarrow \frac{1}{\xi}$, the explicit log tends to the finite $\log \frac{\bar{\mu}^2}{k_{\perp}^2}$ (because $q = \bar{\xi}k \rightarrow k$), and $\xi \mathcal{I}(\xi) \rightarrow \text{const}$ (table in §7). So $g(0)$ exists, the $\log \frac{k_{\perp}^+}{\Lambda}$ is a pure rapidity divergence, and $\left[\frac{1}{\xi} \right]_+$ is exactly the right bookkeeping: the $\log \frac{k_{\perp}^+}{\Lambda}$ goes into the JIMWLK/BK evolution of the target and the $+$ -regulated remainder stays in the NLO impact factor.

At $\xi \rightarrow 1$ — no. There $C_{UV} \rightarrow \frac{1}{\xi}$ but the accompanying function is $\log \frac{\bar{\mu}^2}{\xi^2 k^2} + \bar{\xi} \mathcal{I}$, which *diverges logarithmically* as $\bar{\xi} \rightarrow 0$. $g(0)$ does not exist, the naive $\left[\frac{1}{\xi} \right]_+$ subtracts the wrong thing, and the leftover is the $\frac{1}{2} \log^2$. That is not a defect of the prescription — it is the physical soft-collinear double log.

What to do instead at $\xi \rightarrow 1$. Peel the log off first. Write the $\bar{\xi} \rightarrow 0$ coefficient as $a \log \frac{1}{\bar{\xi}} + b$ and use the two standard distributions

$$\frac{1}{\bar{\xi}} \rightarrow \left[\frac{1}{\bar{\xi}} \right]_+ + \delta(\bar{\xi}) \log \frac{1}{\lambda}, \quad \frac{\log \frac{1}{\bar{\xi}}}{\bar{\xi}} \rightarrow \left[\frac{\log \frac{1}{\bar{\xi}}}{\bar{\xi}} \right]_+ + \frac{1}{2} \delta(\bar{\xi}) \log^2 \frac{1}{\lambda},$$

so the Λ dependence is carried entirely by the two δ terms and the rest is a genuine $+$ -distribution acting on a finite function. This is the same conclusion as in the previous row: the single plus prescription is enough only where the companion function is regular, and here it is regular at exactly one of the two endpoints.

11. What the answer means

- **$-\frac{11}{6}$ is the beta function.** $C_{UV} = \frac{P_{gg}(\xi)}{2N_c}$ and $\int_{\lambda}^{1-\lambda} C_{UV} = 2L - \frac{11}{6}$, with $\frac{11}{6} = \frac{\beta_0}{2N_c}$, $\beta_0 = \frac{11}{3} N_c$ (pure glue). Together with the $\frac{1}{\epsilon_{UV}} + \log \frac{\bar{\mu}^2}{k_{\perp}^2}$ that multiplies it, this is precisely the combination absorbed by $g^2 \rightarrow g^2(\mu)$. That is the “running coupling contribution” your note is named after, and it comes out of this row correctly.
- **No fragmentation function here.** The Wilson lines sit at the *parent* positions x, y, x', w' and there is no $\frac{d\xi}{\xi^2}$ measure with the hard factor evaluated at $k_{\perp}/\xi$; only one gluon is measured and the second is integrated out. The P_{gg} appears as the UV residue of a loop, not as a final-state collinear splitting. Consistent with what the two- $\mathcal{C}$ row gave.
- **One thing to watch.** The product $\frac{1}{\epsilon_{UV}} \times 2 \log \frac{k_{\perp}^+}{\Lambda}$ is a mixed UV $\times$ rapidity divergence. A term like that cannot survive on its own — the μ dependence of a rapidity-divergent piece has nothing to cancel against inside this row. It must cancel against the companion rows of Group I (or be absorbed together with them into the evolution of the coupling inside the JIMWLK kernel). Worth checking explicitly once the other rows are in the same scheme.

12. Checklist of what changed in your note

- (1.1): drop the overall $\frac{v^+}{k^+}$; flip the overall sign. (Your (1.11) already has this right.)
- (1.3): record $q = \bar{\xi}k_{\perp}$, not $k_{\perp}$.
- (1.8): $\left[1 - \frac{r \cdot s}{s^2} \right] \rightarrow \left[1 - \frac{r \cdot s}{2s^2} \right]$.
- (1.9): $\frac{s^2 - s \cdot r}{2(s-r)^2} \rightarrow \frac{s^2 - s \cdot r}{2r^2(s-r)^2}$.
- (1.16)–(1.18): replace by the direct position-space masters of §5.
- (1.20): the pole is $\frac{1}{\epsilon_{UV}} = -\frac{1}{\epsilon}$ in $d = 2 - 2\epsilon$.
- After (1.15): restore the finite tensor term $\pi \bar{\xi} \xi \frac{P \cdot s - 2(P \cdot \hat{k})(s \cdot \hat{k})}{P^2 s^2}$, whose ξ integral is $\frac{1}{6}$.
- (1.21): $I(\xi)$ is *not* uniformly finite; it carries $-\frac{\pi}{\xi} \frac{P \cdot s}{P^2 s^2} \log \frac{1}{\xi}$.
- (1.22): $\log^2 \frac{k_{\perp}^+}{\Lambda} \rightarrow \frac{1}{2} \log^2 \frac{k_{\perp}^+}{\Lambda}$, plus the $(c_0 + c_1)L$.

All algebra verified with `src/row28_check.py`, `src/row28_Tr.py`, `src/row28_full.py`, `src/row28_rem.py`; numbers reproduced in `RESULTS_row28.txt`.